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FULL PAPER

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A theoretical insight into the role of counter anions and their interactions in nitropentaamminecobalt(III) toward linkage isomerism-induced photochemical motion

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Abstract

Density functional theory (B3LYP, B3LYP-D3, wB97XD, M062X, and M06L) and ab initio methods (MP2 and CCSD(T)) in conjunction with 6-31+G(d,p) and LanL2DZ were employed to investigate the interaction energies between $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers and chloride and nitrate in both gas phase and solid state. The nature of the chemical bonding has been analyzed by means of the atoms in molecules, electron density shift, natural bond orbitals, symmetry adapted perturbation theory, and energy decomposition analysis. The electronic structures of the two lowest laying singlet states (S_o and S_1) of $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers were also investigated using CASSCF(6,6) with LanL2DZ and 6-31G(d) basis sets. Our results show that $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers interact more strongly with chloride than nitrate. The structures of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers and their relative stabilities were examined in gas phase and in solid state and confirmed the nitro-complex as the most stable following by a viable intermediate *endo*-complex. Study of the nitro-nitrito linkage isomerization in $[Co(NH_3)_5NO_2](NO_3)Cl$ revealed that anions form strong electrostatic bonds with $[Co(NH_3)_5NO_2]^{2+}$ leading to decrease in an activation energy compared to the $[Co(NH_3)_5ONO]^{2+}$ isomers. A concerted action of ionic interactions and hydrogen bonds are suspected of regulating the isomerization in solid state. Assessment of various DFT methods with respect to CCSD(T) suggests M062X suitable method for $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage-isomerization study. Potential energy surface calculations at the CASSCF/6-31G(d) level of theory shows that the conical intersection (S_1/S_o) might play an important role in the photoisomerization of $[Co(NH_3)_5NO_2](NO_3)Cl$.

KEYWORDS
interaction energies, linkage isomers, nitropentaamminecobalt(III), photosalient effect

1 | INTRODUCTION

Ultraviolet (UV) light irradiation of $[Co(NH_3)_5NO_2](NO_3)Cl$ (**A**) induces an intramolecular linkage isomerization that gives rise to $[Co(NH_3)_5ONO](NO_3)Cl$ (**B**).[1–5] In the solid state, this isomerization transforms only nitro into nitrito, leaving unaffected the lattice positions of all other atoms. This type of intramolecular conversion is autocatalytic or autoinhibitory.[3,4] Naumov et al[1,2] reported that $[Co(NH_3)_5NO_2](NO_3)Cl$ crystals can

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hop off a surface and travel up to several millimeters when exposed to strong UV light. The UV photoisomerization forming $[Co(NH_3)_5ONO]^{2+}$ is suspected to increase local pressure in the crystal, which increase is considered responsible for the observed kinetic motion.[1,2] This photoinduced mechanical motion, also called a photosalient effect,[1] is not intuitively expected for a simple inorganic compound with a discrete ionic structure. This effect is quite intriguing, motivating investigation at the molecular level to know how a miniscule structure perturbation caused by UV irradiation can yield in the crystal structure a concerted action of intermolecular interactions that can result in a release of a collective kinetic energy whose results include jumping, rotation, or fragmentation of the crystal.[1] The origins of the specific physical properties of **A** and **B** are still not well understood. In addition, important questions remain regarding the interaction energies between $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers with nitrate and chloride. Theoretical investigation of **A** and **B** is important for better understanding their chemical bonding and different interactions. The latter may guide researchers in the design of new photosalient materials of same family or others.

Numerous studies have investigated the photoisomerization behavior of $[Co(NH)_5NO_2]^{2+}$ surrounded by inorganic anions.[6–22] The solid state isomerization occurs intramolecularly through the N—O bond[8–10] and is accompanied by an increase in molar volume.[11,13,19] The nitrito isomers of $[Co(NH_3)_5ONO](NO_3)_2$, $[Co(NH_3)_5ONO]Br_2$, $[Co(NH_3)_5ONO]Cl_2$, $[Co(NH_3)_5ONO](NO_3)Cl$, and $[Co(NH_3)_5ONO](ClO_4)$, when stored in the dark in ambient conditions, undergo slow spontaneous transformations into the corresponding nitro isomers.[6–12,21,22] The base dependence and pressure dependence of this linkage isomerization has been studied.[16] Experiments providing detailed information on the synthesis of $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ in various inorganic environments have been reported elsewhere.[17–19]

Our previous study on the isomerization of nitropentaamminecobalt(III) cations in the gas phase revealed that $[Co(NH_3)_5NO_2]^{2+}$ is 1-2 kcal/mol at CCSD(T)/6-31G(d) more stable than its counterpart nitrito isomers.[23] The intramolecular hydrogen bonding (IHB) and the orientation of the atoms within the nitrito group were found to play important roles in the stabilities of the cationic complexes. The nitro/*exo*-nitrito isomerization reaction through the *endo*-$[Co(NH_3)_5ONO]^{2+}$ pathway (viz. nitro → TS1 (38.16 kcal/mol) → *endo*-nitrito → TS2 (9.68 kcal/mol) → *exo*-nitrito) was found to be the most likely mechanism. The *endo*-$[Co(NH_3)_5ONO]^{2+}$ isomer (with ONO in the endohedral orientation) was found to be slightly more stable than its counterpart *exo*-$[Co(NH_3)_5ONO]^{2+}$ isomer (with ONO in the exohedral orientation).[23] There has been continuing interest in the study of $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers due to their potential applications in stimuli-responsive materials, energy harvesting, tunable contact printing, artificial muscles, mechanically tunable optical elements, and switchable reflector elements for projective displays.[21,22] Switching of chemical and physical properties under external stimulus is of great importance in the search for new devices in the nanosciences and nanotechnology.

In the present paper, the geometric structures, interaction energies, relative stabilities, and chemical bonding of $[Co(NH_3)_5ONO](NO_3)Cl$ linkage isomers are studied by means of density functional theory (DFT), second-order Møller-Plesset perturbation *theory* (MP2), coupled cluster theory (CCSD(T)), and complete active space self-consistent field (CASSCF) theory with the purpose to shed some light on the molecular-level cause of these isomers' photomechanical (photosalient) effect. The weak intramolecular O—N···H and N—O···H hydrogen bonding between nitro or nitrito and ammonia hydrogen's is examined in detail, as well as the electrostatic interaction formed by the negatively charged atoms of chloride and nitrate with the polarized H—N bond of ammonia in $[Co(NH_3)_5ONO](NO_3)Cl$ isomers *in vacuo* and in solid state.

The previous studies suggestion that the *endo*-$[Co(NH_3)_5ONO]^{2+}$ isomer may be an intermediate species in the nitro/nitrito isomerization[23] motivated our present study; we assumed that the endohedral orientation of the nitrito group in $[Co(NH_3)_5ONO](NO_3)Cl$ enables more hydrogen bonds to form between O and the H of ammonia in the equatorial region, bringing extra stability to the complex. Furthermore, the orientation of chloride and nitrate anions in $[Co(NH_3)_5NO_2](NO_3)Cl$ and $[Co(NH_3)_5ONO](NO_3)Cl$ is also examined and compared with their positions reported in **A** and **B** crystals.[1,2] The geometries of all structures involved in the nitro conversion mechanism were optimized fully at different levels of theory.

The present work was organized as follows.

1. First, the frontier molecular orbitals and molecular electrostatic potentials (MEPs) of $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers were computed and interpreted as DFT reactivity descriptors *in vacuo*. Ammonia molecules were found to be favorable sites for nucleophilic attacks (eg, anions). The nitro isomer was revealed to have a higher propensity to gain electrons than the nitrito isomer. To test different reactive sites, chloride and nitrate anions were bound to ammonia in different positions (*cis*, axial, and *trans*); the *trans* position was found to be the most favorable position for inorganic anions.
2. Second, binding energies (BEs) between $[Co(NH_3)_5NO_2]^{2+}$ isomers and inorganic anions (chloride and nitrate) were computed using CCSD(T), MP2, B3LYP-D3, M062X, M062L, and ωB97XD, and the interaction energies by symmetry adapted perturbation theory (SAPT2) and energy decomposition analysis (EDA) methods with 6-31+G(d,p) basis set. These interactions were found strong and predominantly ionic. Chloride and nitrate form more stable complexes with $[Co(NH_3)_5NO_2]^{2+}$ than with $[Co(NH_3)_5ONO]^{2+}$.
3. Third, the electron density shift (EDS) and the nature of chemical bonding in $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers were analyzed to elucidate how atoms are held together in the solid state. Hydrogen bonds between nitro (or nitrito) groups and ammonia are weak and can justify the reorientation of atoms in the ONO group, which is at the basis of the observed linkage isomerism. The interactions between chloride and nitrate anions and the $[Co(NH_3)_5NO_2]^{2+}$ isomers are essentially of ionic character and harder to break than hydrogen bonds formed by ammonia. A concerted action of ionic interactions between positive and negative charges of the chemical species in solid linkage $[Co(NH_3)_5ONO](NO_3)Cl$

isomers and IHBs involving nitro and nitrito groups are predicted to play a crucial role in the regulation of the conversion of **A** to **B** initiated by light.

4. Lastly, TD-DFT and CASSCF methods were employed to investigate the photoisomerization. The later arises through ligand-field excited states. The excitation to antibonding d-orbital weakens the Co—NO_2 (Co—ONO) bond and promotes the photochemical reaction. Potential energy surface (PES) investigations show that the conical intersection (CI) between ground and excited singlet states (S_1/S_o) plays an important role in the photoisomerization of $[Co(NH_3)_5NO_2](NO_3)Cl$.

We show through this study that intramolecular conversion of **A** into **B** might passes through a viable *endo* intermediate and that this conversion is mostly regulated by weak IHBs (N—O···H and O—N···H) formed between nitro or nitrito group atoms and ammonia group protons in combination with ionic interactions. The chloride and nitrate form strong ionic bonds that are not expected to be perturbed during the isomerization.

2 | COMPUTATIONAL DETAILS

PES scans performed by incrementing the rotation angle of the nitro and nitrito groups in $Co(NH_3)_5NO_2](NO_3)Cl$ and $[Co(NH_3)_5ONO](NO_3)Cl$ isomers were explored to identify the most favorable orientation of nitro and nitrito groups. The orientations of chloride and nitrate were assigned with respect to the relative positions of the ONO group as *trans* (Figure 1B), axial (Figure 1C), and *cis* (Figure 1D). In the *trans* complex, the inorganic anion and the ONO atoms group are placed on the opposite side while *cis* complex have inorganic anion and ONO atoms group placed both on the same side. All geometries reported in the present paper were fully optimized using CCSD, MP2, B3LYP-D3, M062X, M062L, and ωB97XD methods with the 6-31+G(d,p) basis set. Vibrational frequencies were carried out at the same theory levels, except for CCSD, to ensure that each structure was a true minimum on the PES, and transition states were each characterized by one imaginary vibrational mode. Zero-point energies (ZPEs) were calculated from the harmonic frequencies without correction and represented upper limits to the true anharmonic values.

BEs were derived as electronic energy differences of each complex and its components (namely $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers and their anions), corrected for the basis set superposition error (BSSE) by using the Boys-Bernadi counterpoise[24] and ZPE. Further single-point

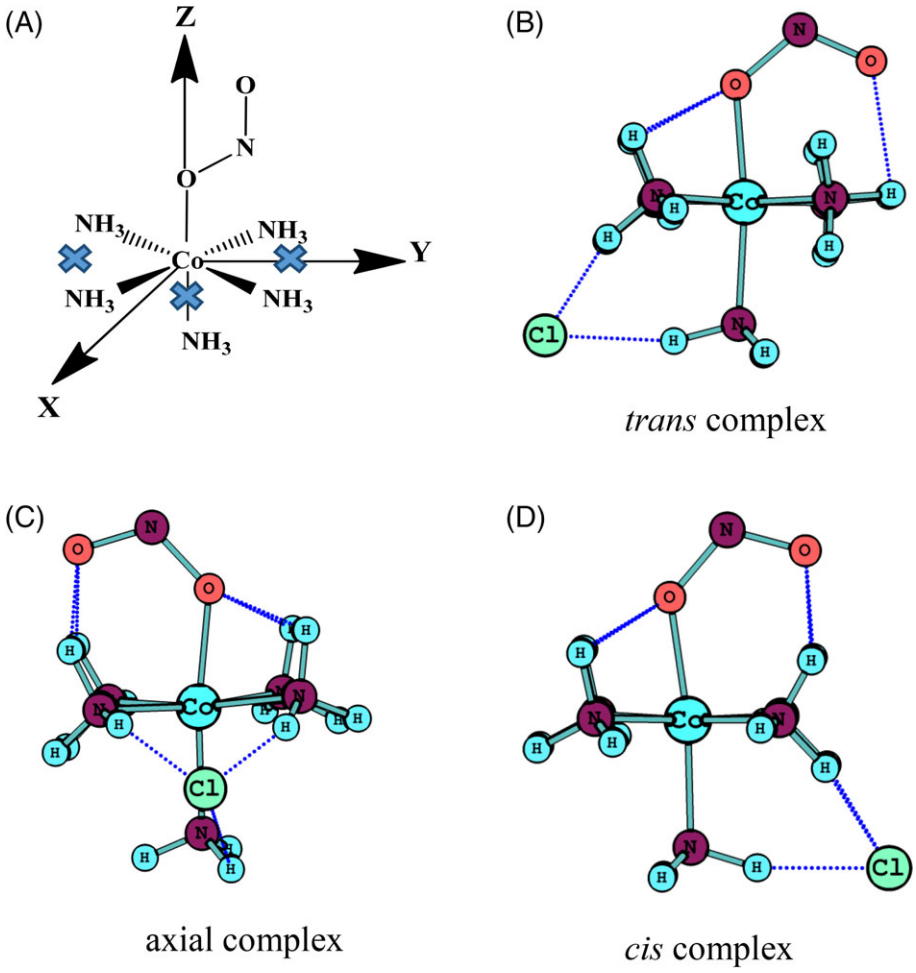

FIGURE 1 Selected positions for chloride (nitrate) used in the initial geometries of the $[Co(NH_3)_5NO_2]Cl^+$ ($[Co(NH_3)_5NO_2]NO_3^+$) complex. The ONO groups are displayed in the *exo* position in (A) and in the *endo* position in (B-D)

energies of the CCSD/6-31+G(d,p) optimized geometries were carried out at CCSD(T)/6-31+G(d,p) level, in attempt to get more refined relative energies and evaluate the performance of DFT methods and MP2 in the characterization of $Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers. MEPs and frontier molecular orbitals were computed as reactivity indices[25–28] to gain insight into the chemical reactivity of $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ isomers. The MEP has been reported as a convenient tool to visualize the anisotropic component of interaction in a complex.[25,26] The theory of atoms in molecules (AIM),[29] the EDS,[26] and natural bond orbitals (NBO)[30] were employed to analyze chemical bonding.

SAPT2, a state-of-the-art method that decomposes interaction energy into charge-transfer, dispersion, electrostatic, exchange-repulsion, and induction, based on the expansion of perturbation theory, was employed to compute the interaction energies.[31] Since the basic unit for a crystalline structure is the unit cell, we have focused our attention on this periodic unit to investigate the structural chemical bonding and energy differences between $Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers. The coordinates of the unit cells of $Co(NH_3)_5NO_2](NO_3)Cl$ and $[Co(NH_3)_5ONO](NO_3)Cl$, denoted respectively as **A** and **B** in the present study, were extracted from Crystallographic Information Files of the experimental crystals[32] to compute solid-state molecular properties. Due to the excessive computational burden arising from the large size of **A** and **B**, the vibrational and electronic spectroscopies were analyzed at M062X/LanL2DZ. All calculations were performed using Gaussian 09[33] and Q-Chem[34] software; Gaussview 5.0,[35] Chemcraft,[36] Jmol 13.0,[37] AIM2000,[38] and Multiwfn[39] software were used for visualization.

3 | RESULTS AND DISCUSSION

3.1 | Reactivity descriptors

The chemical reactivity of a molecular system can be described as an approach of a nucleophile (electrophile) to active sites of respectively maximum (minimum) MEP. To identify electron-rich and -poor regions, MEPs of isolated $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ complexes were plotted (Figure 2); in the figure, electron-rich and -poor regions are shown in red and blue, respectively. The highest negative potential energies in these computed MEP maps were on the nitro and nitrito groups, whereas the most positive potential regions were on the axial hydrogens. Thus, the nitro and nitrito groups are sites having high electron density whereas H atoms are electron-deficient. The nucleophile (eg, NO_3^- or Cl^-) will prefer to transfer its electrons to such regions of low electron density. Further reactivity information was obtained from the frontier molecular orbitals. The highest occupied molecular orbitals (HOMOs) and the lowest unoccupied molecular orbitals (LUMOs) of $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ cationic complexes are given in Figure 3; essentially, the HOMOs (Figure 3A-C) were on NO_2 and ONO only whereas the LUMOs (Figure 3D-F) were on H atoms. The LUMO of a molecular system indicates where an electron can be added. Both MEPs and LUMO maps suggested that the H atoms are preferential sites for nucleophilic attacks (eg, by Cl^- and NO_3^-). Under the Koopmans theorem,[40,41] the LUMO energy value is associated with the vertical electron affinity (EA). Table 1 lists the vertical electron affinities of $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers computed using CCSD(T), MP2, ωB97XD, M062X, B3LYP-D3, and M062L methods with 6-31+G(d,p) basis set. Within this data set, the nitro complex had the highest EA of 6.30 eV at CCSD(T) theory level (6.28 eV at MP2 level), followed by *endo*-nitrito isomer with EA estimated nearly 6.14 eV at CCSD(T) level (6.16 eV at MP2 level) and *exo*-nitrito isomer with EA of 6.11 eV (6.13 eV at MP2 level). Among DFT methods, B3LYP-D3 and M062L yielded the highest (11-12 eV) EA values, whereas ωB97XD and M062X yielded vertical electron affinities of about 9 eV. Thus, the nitro form has a greater propensity to gain electrons than the nitrito form. All DFT methods used in the present study overestimate the electron affinities compare to CCSD(T) and MP2.

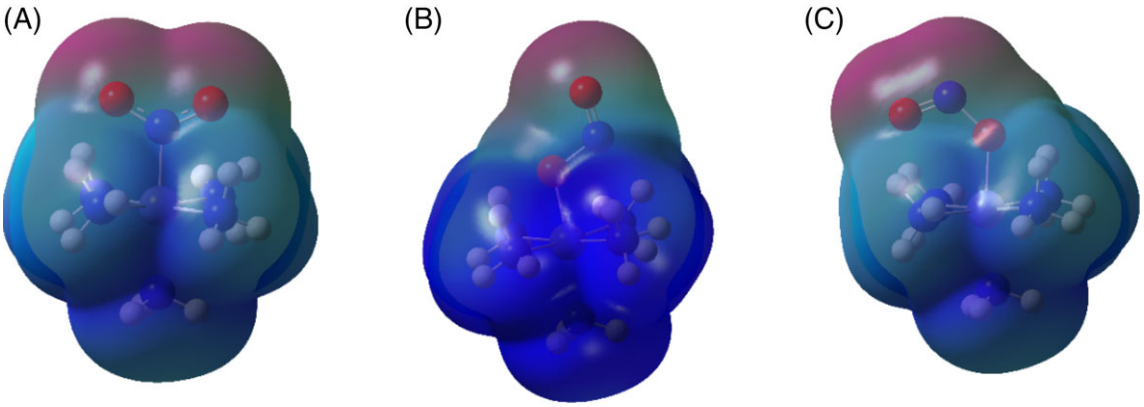

FIGURE 2 Molecular electrostatic potentials (MEPs) of: A, $[Co(NH_3)_5NO_2]^{2+}$, B, *exo*-$[Co(NH_3)_5ONO]^{2+}$, and C, *endo*-$[Co(NH_3)_5ONO]^{2+}$. Red and blue respectively represent regions attractive and repulsive to protons

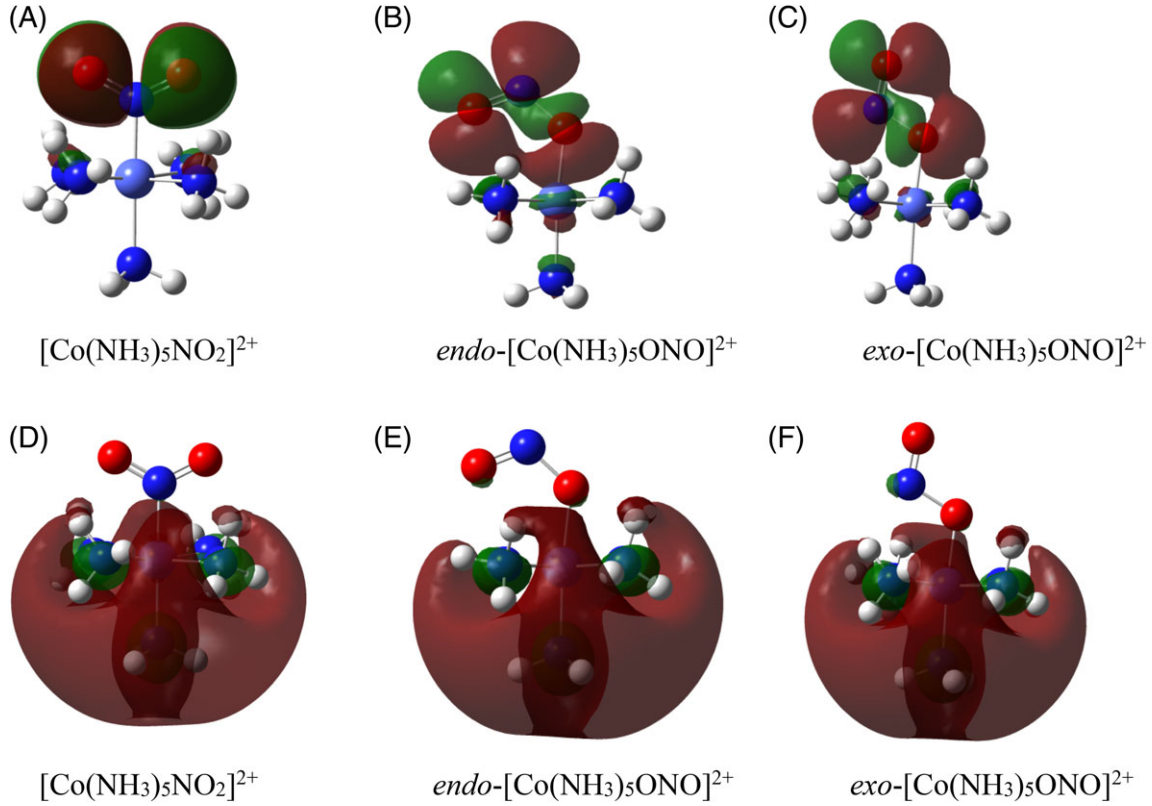

FIGURE 3 Frontier molecular orbitals of $[Co(NH_3)_5NO_2]^{2+}$ isomers: A-C, the highest occupied molecular orbitals (HOMOs) and D-F, the lowest unoccupied molecular orbitals (LUMOs)

TABLE 1 Vertical electron affinities, in eV, of $[Co(NH_3)_5NO_2]^{2+}$ (Co-NO_2), *endo*-$[Co(NH_3)_5ONO]^{2+}$ (*endo*-Co-ONO) and *exo*-$[Co(NH_3)_5ONO]^{2+}$ (*exo*-Co-ONO) cation complexes computed at different theory levels with 6-31+G(d,p) basis set

	ωB97XD	M062X	B3LYP-D3	M062L	MP2	CCSD(T)
Co-NO_2	8.70	9.23	10.84	11.68	6.28	6.30
Endo-Co-ONO	8.70	9.08	10.78	11.55	6.16	6.14
Exo-Co-ONO	8.64	9.07	10.75	11.65	6.13	6.11

3.2 | Structures and energetics of $[Co(NH_3)_5NO_2]Cl^+$ and $[Co(NH_3)_5NO_2]NO_3^+$ linkage isomers

3.2.1 | Equilibrium geometries

Nitrate and chloride anions are electron-rich species that can interact with $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ cations in the cations' electron-poor regions. To identify the best orientations of chloride and nitrate in the lowest-energy atomic structures of the complexes, geometry relaxations were performed of $[Co(NH_3)_5NO_2]Cl^+$ and $[Co(NH_3)_5NO_2]NO_3^+$ linkage isomers with inorganic anions located in the axial, *trans* and *cis* positions (Figure 1). In the present work, *cis* and *trans* positions refer to the positions of nitro and nitrito groups relative to the inorganic anion (chloride or nitrate), which can be either on the same side or opposite side. The axial position refers to structures in which the inorganic anion is along the Co—NH_3 vertical bond. The optimized hydrogen bonds computed with MP2/6-31+G(d,p) are given in Figure 4. The chloride and nitrate interact with $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers through Cl···H—N and O···H—N hydrogen bonds, respectively. The hydrogen bonds formed by chloride with axial NH_3 are shorter compared to those formed with equatorial NH_3, indicating that axial ammonia groups are more amenable to nucleophilic attack. The average hydrogen bond formed by chloride in the nitro complex is around 2.03 Å, slightly shorter than that of the nitrito complex, estimated to be nearly 2.09 Å. Contrastingly, nitrate ions form hydrogen bonds of almost equal average lengths, about 1.74 Å, in the nitro and nitrito complexes. The nitro and nitrito groups are involved in O···H and H···N IHBs with equatorial ammonia molecules. $[Co(NH_3)_5NO_2]Cl^+$, *endo*-$[Co(NH_3)_5ONO]Cl^+$, and *exo*-$[Co(NH_3)_5ONO]Cl^+$ have average IHB lengths of 2.16, 2.36, and 2.30 Å, respectively; the lengths of the respective nitrate forms are 2.25, 2.37, and 2.29 Å. Thus, IHBs tend to be stronger in $[Co(NH_3)_5NO_2]Cl^+$ than in $[Co(NH_3)_5ONO]Cl^+$. According to Muya et al,[23] the difference in IHB in $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers can be explained by the decrease of

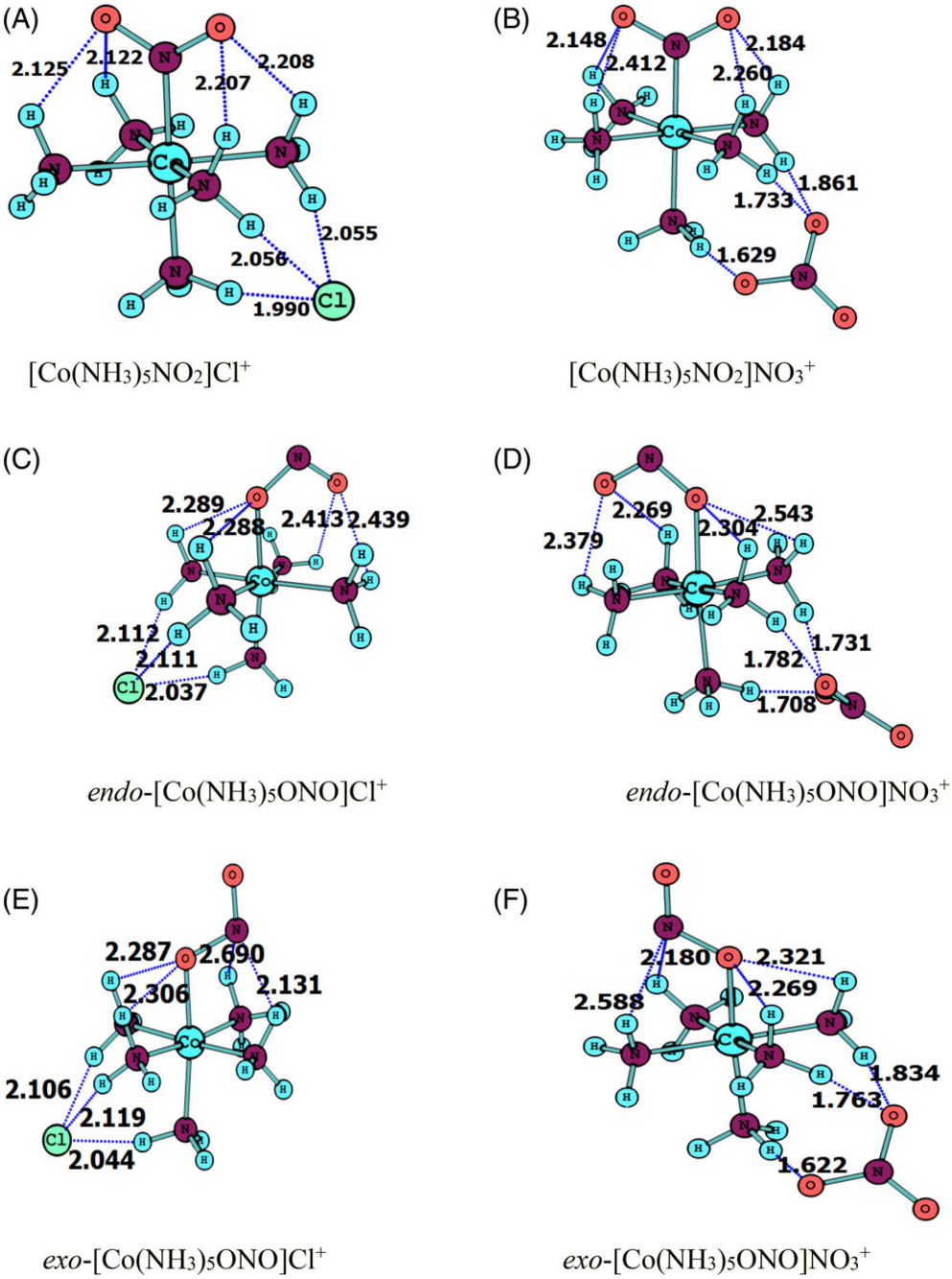

FIGURE 4 Optimized Cl···H, O···H, and N···H hydrogen bonds of A,C,E, *trans*-$[Co(NH_3)_5NO_2]Cl^+$ and B,D,F, *trans*-$[Co(NH_3)_5NO_2](NO_3)^+$ linked isomers computed at MP2/6-31+G(d,p)

electron resonance in the nitrito group compared to the nitro group. Another way to probe the strength of hydrogen bonds in those complexes is to determine the bond lengths of the N—H bonds in the ammonia ligands.

3.2.2 | Relative stabilities

Table 2 lists the relative energies of $[Co(NH_3)_5NO_2]Cl^+$ and $[Co(NH_3)_5NO_2](NO_3)^+$ linkage isomers and their interaction energies. Chloride and nitrate stabilize $[Co(NH_3)_5NO_2]^{2+}$ isomer more than $[Co(NH_3)_5ONO]^{2+}$ isomers and prefer to be in the *trans* orientation. The difference in energy between the axial and *trans* $[Co(NH_3)_5NO_2]Cl^+$ conformers is 0.93 kcal/mol at CCSD(T)/6-31+G(d,p) (1.05 kcal/mol including ZPE and BSSE correction at MP2/6-31+G(d,p) level). The stability of the nitro complexes increases in the following order of orientations of the inorganic anion: *cis*, axial, *trans*. In *endo*- and *exo*-$[Co(NH_3)_5ONO]Cl^+$, the axial and *cis* complexes are 1.04 and 0.65 kcal/mol (computed at CCSD(T)/6-31+G(d,p) level) less stable than their respective *trans* forms. In *exo*-$[Co(NH_3)_5ONO]Cl^+$, all three complexes at MP2 level, that is, with chloride in the *cis*, axial, and *trans* positions, have almost equal probabilities of occurrence; thus there is no orientation preference for nucleophilic attack.

TABLE 2 Relative energies of complexes formed between inorganic anions (chloride, nitrate) and $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ cations, in kcal/mol, computed at CCSD(T)/6-31+G(d,p) and MP2/6-31+G(d,p) values in parenthesis corrected for zero-point energy (ZPE) and basis set superposition error (BSSE)

	Position	Nitro	*Endo*-nitrito	*Exo*-nitrito
Chloride	*cis*	0.00 (0.00)	3.22 (7.85)	4.04 (6.66)
	axial	0.93 (1.05)	3.22 (6.74)	4.04 (6.68)
	trans	0.00 (0.00)	2.18 (6.12)	3.39 (6.50)
Nitrate	*trans*	0.00 (0.00)	0.98 (5.44)	2.36 (5.89)

According to the data listed in Table 3, $[Co(NH_3)_5NO_2]Cl^+$, *endo*-$[Co(NH_3)_5ONO]Cl^+$, and *exo*-$[Co(NH_3)_5ONO]Cl^+$ with chloride in *trans* have BEs of −199.83, −197.41, and −198.02 kcal/mol at CCSD(T) level, respectively, laying in the range of ionic interactions. *Endo*-$[Co(NH_3)_5ONO]Cl^+$ and *exo*-$[Co(NH_3)_5ONO]Cl^+$ are 3.22 and 4.04 kcal/mol less stable than $[Co(NH_3)_5NO_2]Cl^+$ at the same theory level. M062X relative energies and BEs are approaching CCSD(T) compare to MP2 and M062L. The chloride binds more strongly to $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers than nitrate does. $[Co(NH_3)_5NO_2]Cl^+$ isomers are 9 kcal/mol more stable than $[Co(NH_3)_5NO_2]NO_3^+$ based on computed CCSD(T) BEs. All the three $[Co(NH_3)_5ONO]NO_3^+$ linkage isomers have almost similar BEs of about −189 kcal/mol at CCSD(T)/6-31+G(d,p) level which is in prefect agreement with the computed M062L BEs of −188 kcal/mol. As the case of $[Co(NH_3)_5ONO]Cl^+$ isomers, *endo* and *exo*-$[Co(NH_3)_5ONO]NO_3^+$ linkage isomers are 0.81 and 2.24 kcal/mol at CCSD(T) level less stable than their counterpart nitro (1.46 and 1.58 kcal/mol at M062X level). M062L and MP2 have tendency to overestimate the relative energies of $[Co(NH_3)_5ONO]Cl^+$ and $[Co(NH_3)_5ONO]NO_3^+$ linkage isomers linkage. The CCSD(T)/6-31+G(d,p) calculations show that $[Co(NH_3)_5NO_2]^{2+}$ is 0.81 and 2.24 kcal/mol more stable than the *endo*-$[Co(NH_3)_5ONO]^{2+}$ and *exo*-$[Co(NH_3)_5ONO]^{2+}$. This can suggest that the chloride compared to nitrate tends to stabilize more the nitro form than the nitrito.

The total interaction energies as well as their components (eg, electrostatic, induction, repulsion, and dispersion energy) for each complex were computed using the symmetry-adapted perturbation theory (SAPT2). The overall set of different contributions of interaction energies of $[Co(NH_3)_5NO_2]X^+$ (X=Cl, NO_3) linkage isomers are listed in Table 4. The interaction energies between chloride and *exo*-$[Co(NH_3)_5ONO]^{2+}$, *endo*-$[Co(NH_3)_5ONO]^{2+}$, and $[Co(NH_3)_5NO_2]^{2+}$ computed using SAPT2 protocol were found nearly −196.63, −197.44 and −199.20 kcal/mol at B3LYP/6-31+G(d,p) (−190.91, −191.96, −194 kcal/mol at HF/6-31+G(d,p)) and that of *exo*-$[Co(NH_3)_5ONO](NO_3)^+$, *endo*-$[Co(NH_3)_5ONO](NO_3)^+$, and $[Co(NH_3)_5NO_2](NO_3)^+$ were estimated nearly −191.60, −193.61 and −193.62 kcal/mol at B3LYP level (−187.69, −190.18, and −191.01 kcal/mol at HF level). These computed SAPT2 interaction energies approach the CCSD(T) BEs listed in Table 3. The dispersion energy in linkage isomers formed by nitrate differed from those of the

TABLE 3 Relative energies and binding energies in parentheses between chloride and nitrate with $[Co(NH_3)_5NO_2]^{2+}$ isomers in kcal/mol, of the most stable isomers computed at different theory levels with 6-31+G(d,p) basis set

Anions	Methods	Nitro	*Endo*-nitrito	*Exo*-nitrito
Chloride	M062X	0.00 (−199.26)	0.84 (−196.08)	2.56 (−197.21)
	M062L	0.00 (−198.34)	6.39 (−195.43)	9.69 (−197.81)
	MP2	0.00 (−199.33)	6.12 (−199.23)	6.50 (−197.92)
	CCSD(T)	0.00 (−199.83)	3.22 (−197.41)	4.04 (−198.02)
Nitrate	M062X	0.00 (−191.22)	1.46 (−190.35)	1.58 (−190.15)
	M062L	0.00 (−188.37)	3.48 (−188.11)	9.16 (−188.74)
	MP2	0.00 (−193.25)	5.44 (−193.29)	5.89 (−191.47)
	CCSD(T)	0.00 (−189.41)	0.81 (−189.24)	2.24 (−189.29)

TABLE 4 Electrostatic, induction/polarization, and repulsion energies, in Hartrees, and dispersion energies in kcal/mol, as well as interaction energies in kcal/mol for each complex calculated using the symmetry adapted perturbation theory (SAPT2) scheme at HF/6-31+G(d,p) (in parentheses: using B3LYP/6-31+G(d,p))

Complex	Electrostatic	Exchange-electrostatic	Induced	Exchange-induction	dispersion	Interaction energy
$[Co(NH_3)_5NO_2]NO_3^+$	−208.58 (−206.75)	57.10 (68.50)	−37.02 (−65.51)	13.62 (26.77)	−16.29 (−16.29)	−191.01 (−193.62)
Endo-$[Co(NH_3)_5ONO]NO_3^+$	−207.56 (−206.64)	55.87 (67.00)	−35.67 (−64.19)	13.15 (26.18)	−15.97 (−15.97)	−190.18 (−193.61)
Exo-$[Co(NH_3)_5ONO]NO_3^+$	−205.97 (−205.77)	57.13 (68.22)	−36.33 (−64.52)	26.46 (13.47)	−15.98 (−15.98)	−187.69 (−191.60)
$[Co(NH_3)_5NO_2]Cl^+$	−221.77 (−220.02)	66.29 (74.13)	−55.22 (−94.31)	29.12 (53.84)	−12.82 (−12.82)	−194.44 (−199.20)
Endo-$[Co(NH_3)_5ONO]Cl^+$	−215.01 (−214.11)	58.01 (65.24)	−49.21 (−85.37)	25.92 (48.46)	−11.67 (−11.67)	−191.96 (−197.44)
Exo-$[Co(NH_3)_5ONO]Cl^+$	−213.80 (−213.52)	57.60 (64.82)	−48.87 (−84.53)	25.78 (48.22)	−11.62 (−11.62)	−190.91 (−196.63)

corresponding chloride forms by nearly 4 kcal/mol and contributed about 6% of the total attractive interaction forces; electrostatic forces are the main contributors to the attractive and repulsive energies occurring in the complexes, estimated at nearly 80%. Hence, these interactions are mostly electrostatic in nature. The inorganic anions interact with $[Co(NH_3)_5ONO]^{2+}$ isomers through polarized N—H bonds. The electron densities and Laplacians at the N—H···Cl and O···N—H bond critical points (bcps) in $[Co(NH_3)_5NO_2]X^+$ (X=Cl^-, NO_3^-) linkage isomers are listed in Supporting Information Table S1. These electron density topological properties provide information on the hydrogen bonds (H-bonds) formed by chloride and nitrate with $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers. The hydrogen bonds formed by nitrate appear to be slightly stronger than those formed by chloride based on the magnitude of the electron densities and their Laplacians at various H···Cl and O···N—H bcps.

3.3 | Structures and energetics of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers *in vacuo*

3.3.1 | Equilibrium geometries

Supporting Information Figure S1A shows a superimposition of the PES of $Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers obtained at M062X/6-31G (d) after rotation of nitro and nitrito groups around Co—N and Co—O bond. All global minima on PES have chloride in *trans* position whereas geometric structures with chloride oriented in *cis* and axial were identified as local minima. The PES of $[Co(NH_3)_5NO_2](NO_3)Cl$ is the lowest followed by that of the *endo*-$[Co(NH_3)_5ONO](NO_3)Cl$. The overall geometric structures of the lowest complexes formed by the interactions of chloride and nitrate with $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers optimized at MP2/6-31+G(d,p) are shown in Figure 5. $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers had longer hydrogen bonds compared to $[Co(NH_3)_5NO_2]Cl^+$ and $[Co(NH_3)_5NO_2]NO_3^+$ isomers (Figure 4). The N(axial)-Co-N(nitro) interatomic angles were about 176° whereas the N(axial)-Co-O(nitrito) angles were about 172°. The average Cl···H—N bonds in nitro, *endo*- and *exo*-nitrito complexes were about 2.114, 2.147, and 2.180 Å long, respectively, and the corresponding O···H—N bonds were about 1.827, 1.867, and 1.876 Å.

3.3.2 | Relative stabilities and interaction energies

Supporting Information Table S2 compares the relative energies of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers computed at CCSD(T) and MP2 levels with 6-31+G(d,p), for chloride and nitrate positions of either *trans* or *cis*. According to this analysis, there appeared to be no preference in the orientation of chlorine and nitrate. Table 5 lists the relative energies and BEs of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers computed using CCSD(T), MP2, and various DFT methods. The nitro complex had the lowest energy, differing from the *endo*- and *exo*-nitrito complexes by 3.54 and 3.70 kcal/mol at CCSD(T)/6-31+G(d,p) level. MP2 and DFT methods used in the present study fail to predict that *exo*- and *endo*-nitrito complexes

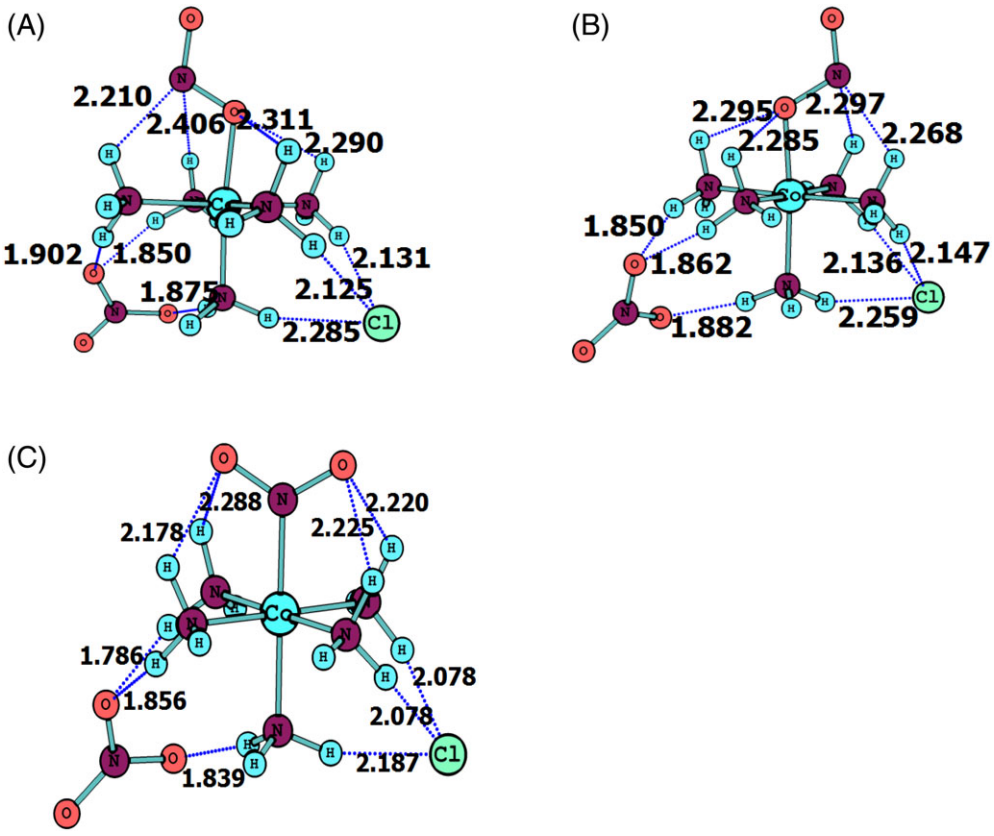

FIGURE 5 Hydrogen bonds distances in *exo*- $[Co(NH_3)_5ONO](NO_3)Cl$ with Cl in trans (A) or cis (B) and in $[Co(NH_3)_5NO_2](NO_3)Cl$ (C)

have almost same relative energy in gas phase. Chloride and nitrate bind more strongly to $[Co(NH_3)_5NO_2]^{2+}$ (BE of −323.09 kcal/mol computed at CCSD(T)/6-31+G(d,p)) than $[Co(NH_3)_5ONO]^{2+}$ isomers (BE of −320.35 and −321.63 kcal/mol computed at CCSD(T)/6-31+G(d,p)).

By means of the absolutely localized molecular orbital analysis implemented in Q-Chem Quantum Chemistry software, the interaction energies between inorganic anions (chloride and nitrate) and $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers were separated into chemically distinct components, namely frozen density (FRZ), polarization (POL), and charge transfer (CT) as follows.

$$\Delta E_{\text{in}} = \left(\Delta E_{\text{FRZ}} + \Delta E_{\text{POL}} + \Delta E_{\text{CT}} + \Delta E_{\text{HF-BSSE}}\right) + \Delta E_{\text{CORR}}$$

All terms in parentheses in this equation were computed at the HF/6-31+G(d,p) level. The last term represents the high correction energy and was calculated as a BSSE-corrected correlation energy at the MP2 level. Table 6 lists the values of each component of the interaction energies. The interaction energies quantified by this EDA approach (ca −325 and −331 kcal/mol) lay in the same range as the BEs reported in Table 5. The interaction energy is more important in the nitro complexes (calculated about −331.10 kcal/mol) than in the nitrito complexes. *Endo*- and *exo*-nitrito complexes have equal interaction energies of −325.36 and −325.79 kcal/mol, respectively. Analysis of the charge transfers in those complexes showed that chloride was responsible for most of the charge transfer energy. In the nitro complex, for instance, chloride contributed nearly 75% of the total charge transfer energy. The polarization energy was estimated to be about −30 kcal/mol in those linkage isomers, slightly lower in magnitude than the sum of charge transfer energy and delocalization energy (CT + DEL), evaluated to be around −34 kcal/mol in the nitro complex and −32 kcal/mol in the nitrito complex.

3.3.3 | Linkage isomerization of $[Co(NH_3)_5(NO_2)](NO_3)Cl$ in gas phase

The *endo*-$[Co(NH_3)_5ONO](NO_3)Cl$ can be suspected as a possible intermediate, but has not yet been captured and stabilized experimentally. Muya et al[23] suggested that the nitrito/*exo*-nitrito isomerization reaction occurs via the $[Co(NH_3)_5NO_2]^{2+} \rightarrow [Co(NH_3)_5NO_2]^{2+}$ (TS1) → [*endo*-$Co(NH_3)_5ONO]^{2+} \rightarrow [Co(NH_3)_5NO_2]^{2+}$ (TS2) → [*exo*-$Co(NH_3)_5ONO]^{2+}$ reaction pathway. The transformation of $[Co(NH_3)_5NO_2]^{2+}$ into [*endo*-$Co(NH_3)_5ONO]^{2+}$ was found to be thermodynamically controlled whereas that of [*endo*-$Co(NH_3)_5ONO]^{2+}$ to [*exo*-$Co(NH_3)_5ONO]^{2+}$ was reported to be kinetically controlled. However, in the case of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers, the barrier between the *exo*- and *endo*-nitrito forms *in vacuo* is about 2.67 kcal/mol at CCSD(T)/6-31+G(d,p) (5.37 kcal/mol in the case of $[Co(NH_3)_5ONO]^{2+}$ conversion) and that between nitro and *endo*-nitrito on the ground state is about 22.98 kcal/mol (34.05 kcal/mol for $[Co(NH_3)_5NO_2]^{2+}$ conversion) at the same theory level with respect to the nitro form (Table 7, Figure 6). The barrier energies and energetic profiles of the linkage isomerization computed at different theory levels

TABLE 5 Relative energies between $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers (in parentheses: binding energies), in kcal/mol, computed at CCSD(T), MP2, M062X, M062L, ωB97XD, and B3LYP-D3. Chloride is in the *trans* position

Methods	Nitro	*Endo*-nitrito	*Exo*-nitrito
M062X	0.00 (−322.06)	1.26 (−318.48)	2.14 (−320.44)
M062L	0.00 (−318.82)	6.04 (−316.26)	8.54 (−319.43)
ωB97XD	0.00 (−327.70)	2.16 (−327.12)	4.76 (−327.12)
B3LYP-D3	0.00 (−329.92)	2.63 (−330.02)	5.84 (−329.96)
MP2	0.00 (−326.32)	7.08 (−321.64)	7.61 (−322.14)
CCSD(T)	0.00 (−323.09)	3.54 (−320.35)	3.70 (−321.63)

TABLE 6 Energy decomposition analysis, in kcal/mol, of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers computed at HF/6-31+G(d,p)

Components[a]	Nitro	*Endo*-nitrito	*Exo*-nitrito
FRZ	−264.09	−260.71	−262.65
POL	−30.07	−29.21	−29.01
DEL	−18.01	−17.24	−16.68
CT	−16.29	−15.56	−15.03
BSSE	1.72	1.67	1.66
Higher order corr.	−4.42	−4.21	−4.07
Int. energies	−331.10	−325.36	−325.79

[a]FRZ, POL, DEL, and CT respectively represent frozen density, polarization, delocalization, and charge transfer energies.

TABLE 7 Relative energies and activation energies (in kcal/mol) of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers and that of $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers in parenthesis computed at different theory levels with 6-31+G(d,p) basis set in gas phase

	Nitro	TS1	*Endo*	TS2	*Exo*
M062x	0.00 (0.00)	29.63 (36.59)	1.26 (−2.33)	9.71 (6.83)	2.14 (0.51)
M062L	0.00 (0.00)	32.91 (39.58)	6.04 (3.48)	24.35 (19.24)	8.54 (9.16)
CCSD(T)	0.00 (0.00)	22.98 (34.05)	3.56 (0.81)	6.22 (6.18)	3.70 (2.24)
MP2	0.00 (0.00)	21.54 (39.69)	8.45 (3.77)	11.59 (5.56)	7.05 (2.88)
ωB97XD	0.00 (0.00)	32.38 (40.89)	8.19 (2.32)	4.80 (10.38)	9.88 (4.39)

such as M062L, M062X, MP2, and ωB97XD with 6-31+G(d,p) basis sets, as shown in Figure 6B and Table 7, indicate that MP2 yields an energetic profiles close to CCSD(T) results. Among DFT methods used, one can notice a good performance of M062X. This assessment provides useful guidance for choosing appropriate quantum chemistry approach for $[Co(NH_3)_5NO_2]NO_3Cl$ linkage isomers.

Chloride and nitrate might act as catalysts by forming strong electrostatic bonds with the $[Co(NH_3)_5NO_2]^{2+}$ isomers, thereby lowering the energy of activation of the isomerization of the $[Co(NH_3)_5NO_2](NO_3)Cl$ complex compared to the $Co(NH_3)_5NO_2^{2+}$ isomerization (Figure 6A). The small difference energies barrier between *endo*- and *exo*-isomers combined with the small relative energies between them can lead to the suggestion that the two isomers might be in equilibrium in gas phase. The $[Co(NH_3)_5NO_2](NO_3)Cl$ transition states between nitro- and *endo*-complex and that connecting *endo*- and *exo*-complex, denoted respectively as TS1-c and TS2-c, are given in Figure 6 and are each characterized by one vibrational imaginary mode of *i430* cm^{-1} and *i199* cm^{-1} at ωB97XD/6-31+G(d,p).

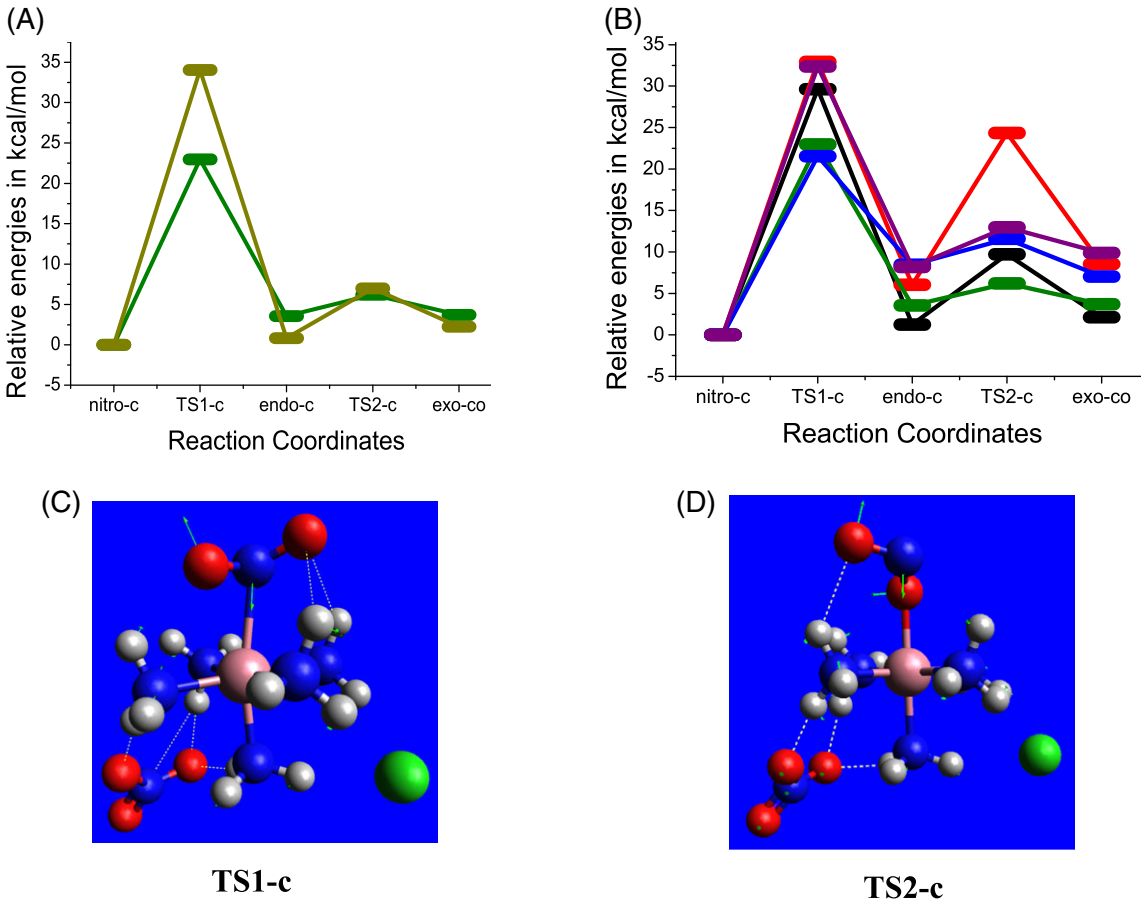

FIGURE 6 A, Reaction pathway of the conversion of $[Co(NH_3)_5NO_2](NO_3)Cl$ into *exo*-$[Co(NH_3)_5NO_2](NO_3)Cl$ in green and that of $[Co(NH_3)_5NO_2]^{2+}$ into *exo*-$[Co(NH_3)_5ONO]^{2+}$ in dark yellow computed at CCSD(T)/6-31+G(d,p) in gas phase. B, Reaction pathway of the conversion of $[Co(NH_3)_5NO_2](NO_3)Cl$ computed at different levels: CCSD(T) (green), M062L(red), M062X (black), MP2 (blue), and ωB97XD (purple). C, Transition state connecting nitro and *endo*-nitrito (TS1, *i430* cm^{-1}) and D, that laying between *endo*- and *exo*-nitrito (TS2, *i199* cm^{-1}), computed at ωB97XD/6-31+G(d,p) *in vacuo*

3.3.4 | Electron density shift

The EDS was computed as the electron density difference between the complex and its components (Figure 7) to investigate how the electron density of the $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers and that of the chloride and nitrate anions adjust in $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers. The green and blue regions depicted in Figure 7 respectively represent losses and gains of electrons. The O···H and N···H IHBs involving NO_2 and ONO atoms groups were evident as electron gains around bridging protons and electron losses on O and N. A different picture was observed for the maps of electron density shift around the Cl···H—N and O···H—N hydrogen bonds occurring in $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers, characterized by gain of electron density on Cl and on the O of the nitrate, and loss of density on the neighboring H. The Cl and the O of the nitrate in those complexes involved in the interaction with ammonia were polarized by the cationic $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers. This indicates that the shared electrons are not equally attracted and that the bonds formed between chloride or nitrate and ammonia are predominantly ionic in character. NBO calculations indicated that one 4s, three 3p, and two 3d orbitals of Co(III) centered within the octahedron hybridizes to form sp^3d^2 orbitals and contributed nearly 15% to the overall Co—NH_3 bond formation. Because the N of NH_3 has a lone pair of electrons, it donates its lone pair to an empty orbital to pair the empty orbital of Co[23] and withdraws electrons from H. The latter becomes more acidic and can interact with electron-rich species such as chloride, nitrate, or ONO atom groups to compensate for its electron deficiency.

3.3.5 | Photoisomerization of $[Co(NH_3)_5NO_2](NO_3)Cl$

CASSCF(6,6) combined with LANL2DZ and 6-31G(d) basis sets were employed to investigate the photoisomerization of $[Co(NH_3)_5NO_2](NO_3)Cl$. These CAS orbitals comprise bonding and antibonding orbitals and mostly reside on the metal (Supporting Information Figure S2). The geometrical parameters of the low-laying singlet states (S_o and S_1) optimized at CASSCF(6,6)/6-31G(d) of $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers are given in Supporting Information Table S3. Optimization of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers in ground and first excited singlet state at CASSCF (6,6)/6-31G(d,p) reveals an elongation of Co—NO_2 and Co—ONO bonds of 0.114 and 0.016 Å, respectively, in the first excited singlet state geometries with respect to their corresponding ground state geometries. Co—NH_3 bonds in excited states lie in between 1.949-2.183 Å in nitro-complex and 1.964-2.137 Å in nitrito-complex and appear shortened and elongated compare to ground state geometries (1.993-2.002 Å for nitro

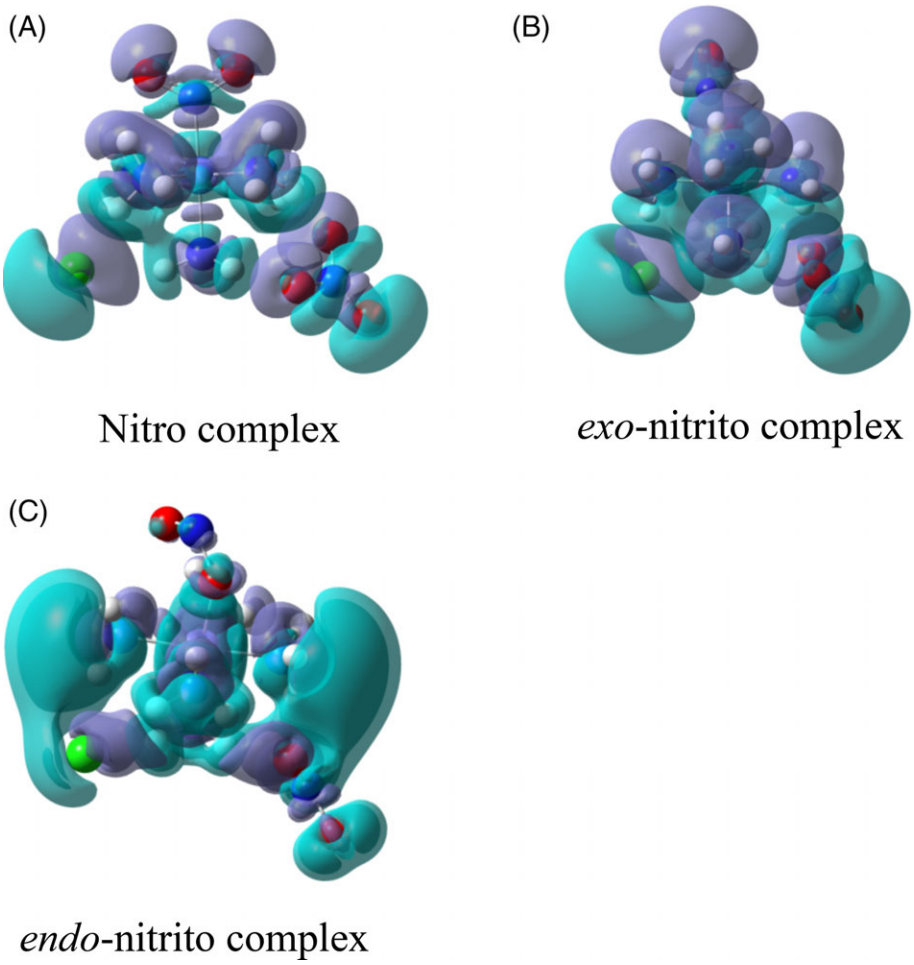

FIGURE 7 Electron density shifts occurring within $[Co(NH_3)_5NO_2](NO_3)Cl$ complexes. Blue and green regions respectively indicate gains and losses of electron density

and 1.990-2.000 Å for nitrito-complex). The number of electrons occupying the six active molecular orbitals of the ground and first excited singlet states of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers computed by means of NBO approach at CASSCF(6,6)/6-31G(d) are given in Supporting Information Table S4. The CASSCF(6,6)/6-31G(d) configuration of the ground states of these isomers is $(72a)^2(73a)^2(74a)^2$ and contributes by about 95% to the total wave function. The first excited singlet states of nitro- and nitrito-complexes are both characterized with strong multi-reference character (Supporting Information Tables S4 and S5). The energy gap between the ground state (S_o) and the first excited singlet state (S_1) of the nitro-compound is about 50 kcal/mol at CASSCF(6,6)-MP2 at 6-31G(d). The electronic energies difference including ZPEs (S_1-S_o energy gaps) of $[Co(NH_3)_5NO_2](NO_3)Cl$, endo-$[Co(NH_3)_5ONO](NO_3)Cl$ and *exo*-$[Co(NH_3)_5ONO](NO_3)Cl$ computed at TD-B3LYP/6-31G(d) were found about 58.12, 46.69, and 46.53 kcal/mol, respectively. The electronic structure of the excited singlet state shows clearly that one electron from 74a active molecular orbital (HOMO) of the ground state (S_o) has been promoted to 76a molecular orbital of nitro-complex (75a MOs for the *exo*-nitrito-complex). A close inspection of these active MOs reveals that 76a and 75a MOs are antibonding of d-type mostly localized on metal and have similar shapes, while the HOMO (74a) has a bonding character with high contribution on both metal and NO_2 ligand (Supporting Information Figure S2). The excitation to the antibonding d-orbital weakens the Co—N bond or Co—O bond and is suspected to promote photochemical reaction. TD-B3LYP/6-31G(d) calculations show that the $[Co(NH_3)_5NO_2](NO_3)Cl$ in the ground state is promoted to the first excited singlet state after photon absorption around 400 nm resulting from NO_2 ligand to Co metal charge transfer (LMCT). The CI between S_o and S_1 of linkage isomer was searched from the Franck-Condon geometry of the $[Co(NH_3)_5NO_2](NO_3)Cl$ compound using state averaged CAS orbitals with LanL2DZ basis set. The energy difference between the two states (S_o-S_1 energy gap) at the CI was estimated to be nearly 0.027 kcal/mol at CASSCF(6,6)/LanL2DZ. The geometry of $[Co(NH_3)_5NO_2](NO_3)Cl$ at this CI is somewhat reminiscent of the CI geometry $(S_1/S_o)_{CI}$ of 4-nitromidazole reported by Yu et al.[42] The PES built from the CI (S_1/S_o) of $[Co(NH_3)_5NO_2](NO_3)Cl$ plotted by means of intrinsic reaction coordinate (IRC) method at CASSCF (6,6)/Lanl2DZ is provided in Supporting Information Figure S3. The IRC performed from this CI following the downhill path for the lowest of the two roots (paths) leads to the Franck-Condon geometry of the nitro complex and *endo*-nitrito complex (Supporting Information Figure S4), as the case of the photoinduced isomerization in 4-nitroimidazol.[42] Thus, one can suggest the photoirradiation process of nitro-complex to proceed as follows: first the $[Co(NH_3)_5NO_2](NO_3)Cl$ in the ground state is promoted to the first excited singlet state after photon absorption (LMCT), then relaxes to the CI (S_o/S_1) and afterward follows one downhill direction toward either $[Co(NH_3)_5NO_2](NO_3)Cl$ or endo-$[Co(NH_3)_5ONO](NO_3)Cl$ ground singlet state. The *endo*-complex can therefore transforms at the ground state into *exo*-$[Co(NH_3)_5NO_2](NO_3)Cl$ owing to the small barrier energy and relative electronic energy between them (Figure 6).

3.4 | Analysis of $[Co(NH_3)_5NO_2](NO_3)Cl$ and $[Co(NH_3)_5ONO](NO_3)Cl$ in solid state

3.4.1 | Comparative study of stability and chemical bonding

The crystal structures of $[Co(NH_3)_5NO_2](NO_3)Cl$ and $[Co(NH_3)_5ONO](NO_3)Cl$, denoted as **A** and **B** in accordance with Refs. [12,13], were examined to elucidate the influence of the nitrate and chloride upon $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ in the solid state. The geometries of these crystals belong to the C_i symmetry group according to Schoenflies notation and are depicted in Supporting Information Figure S5A,B. The crystal packings for **A** and **B** as depicted in the figure comprise four $[Co(NH_3)_5NO_2]^{2+}$ isomers interacting with chloride and nitrate anions aligned diagonally into the plane of the paper, and can be assimilated to a tetramer of $[Co(NH_3)_5NO_2](NO_3)Cl$ isomers if one adds the two missing nitrate anions close to axial NH_3 in *trans* orientation to preserve the molecular C_i symmetry (Figure 8). The positions of nitrate and chloride in the single crystal X-ray structures[32] are in good agreement with our predicted MEP (Figure 2) and computed BEs that suggest the anions to bind strongly to the axial ammonia in *trans* position. However, packed $[Co(NH_3)_5NO_2]^{2+}$ and *exo*-$[Co(NH_3)_5ONO]^{2+}$ cations in **A** and **B** solid are reversely rotated around chloride to keep the centrosymmetric space group. Two sites can be distinguished following the position of chloride: *axial* and *trans*. These unit cells were optimized at M062X/6-31+G(d,p) and characterized by real frequencies to ensure that there are minima on the potential energies. Furthermore, the *exo*-nitrito group in the experimental unit cell of crystalline structure **B**[32] was converted in *endo*-nitrito in C_i symmetry, to evaluate its relative stability with respect to the nitro (**A**) and *exo*-nitrito (**B**). The single point energies difference between the experimental geometries of **A** and **B** computed at HF/LanL2DZ shows **A** to be 20.32 kcal/mol more stable than **B**, while M062X/6-31+G(d,p) geometry optimization reveals that **A** lies 10 and 0.15 kcal/mol (25.98 and 11.90 kcal/mol at B3LYP-D3/6-31+G(d,p)) below **B** and the *endo*-form. Hence, this *endo*-form appears to be a possible viable intermediate structure in solid state upon linkage isomerization process.[23]

Figure 9 shows the superimposition of IR and UV spectra of **A**, **B**, and *endo*-form structures at M062X/LanL2DZ. Table S6 compares experiment and theoretical values of the main modes involving N—O bonds. The IR spectroscopic study of $[Co(NH_3)_5(NO_2)](NO_3)Cl$ linkage isomers in solid state at M062X/LanL2DZ shows the computed deformation and stretching N—O modes differ slightly between isomers and approach experimental IR frequencies.[15] **B** isomer is characterized by a vibrational mode around 1576 cm^{-1} which is absent in **A** and *endo*-form (Figure 9A). The band appearing at 1032 cm^{-1} in the computed IR spectrum of *exo*-$[Co(NH_3)_5ONO](NO_3)Cl$ was assigned to the stretching vibration of N—O bonds attached to metal. However, the UV-Vis spectra of B and its *endo*-form as depicted in Figure 9B are quite similar and differ with that of **A** isomer. The latter is characterized by a second maximum observed around 700 nm which does not appear in nitrito isomers UV-Vis spectra belonging to ligand-field transition signature.

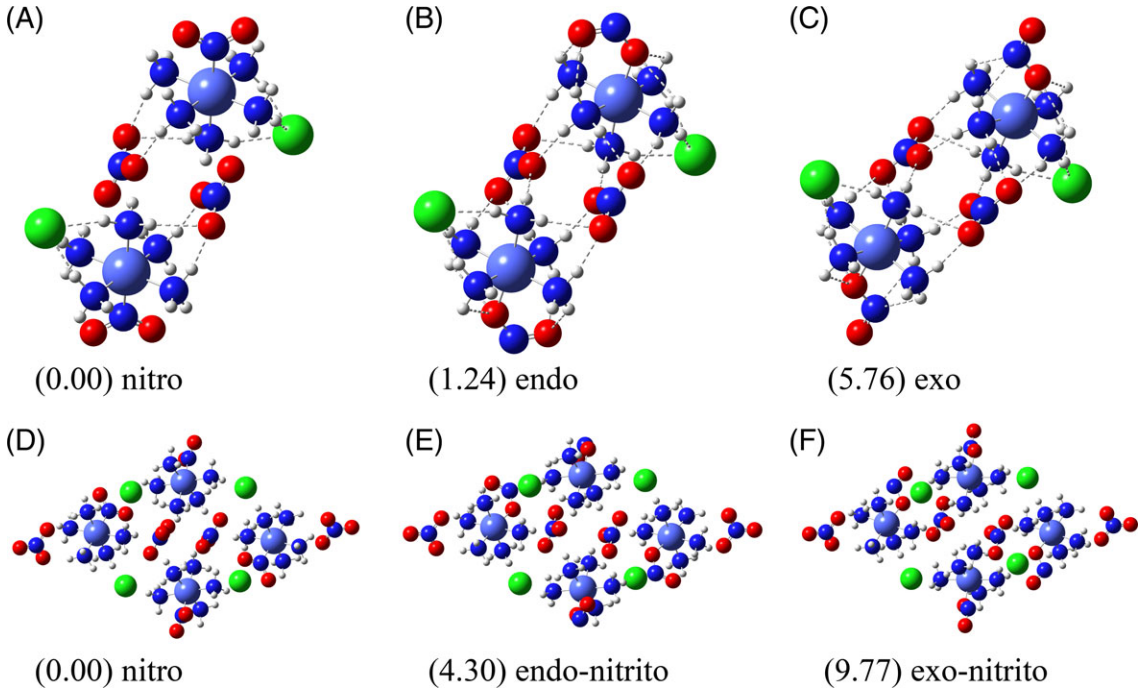

FIGURE 8 Relative energies in kcal/mol of A-C, dimers and D-F, tetramers of $[Co(NH_3)_5NO_2]$ $(NO_3)Cl$ linkage isomers computed at M062X/6-31+G(d,p)

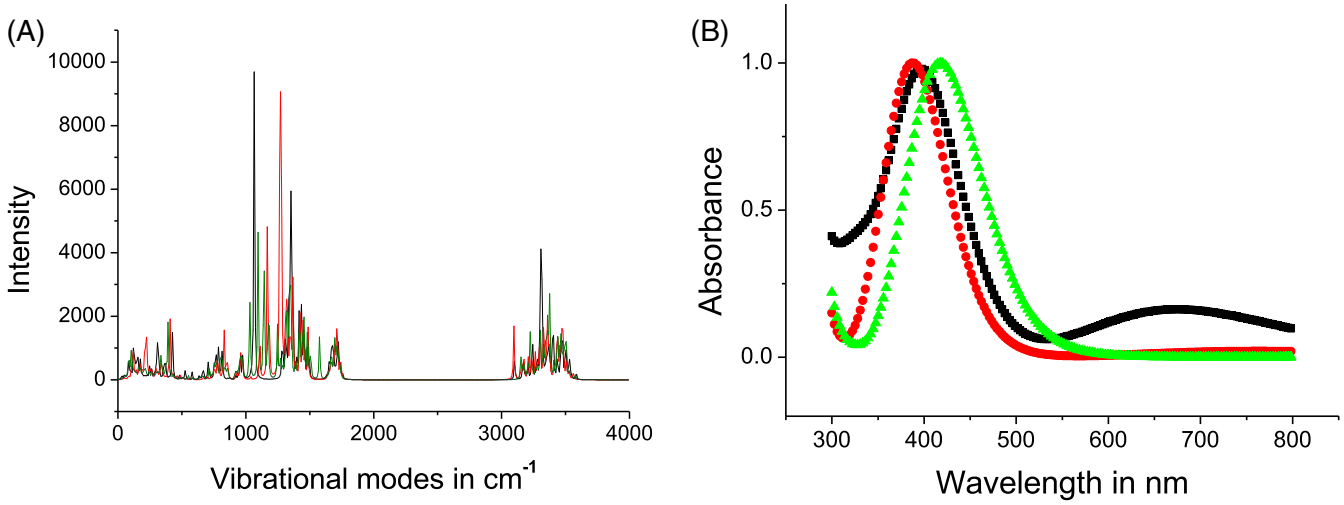

FIGURE 9 Superimposition of A, IR and B, UV-Vis spectra of optimized **A** (black), **B** (green) and *endo*-form (red) structures computed at M062X/6-31+G(d,p). UV absorbances are normalized

The optimized M062X/6-31+G(d,p) and experimental structure parameters (bond lengths and bond angles) of the unit cells are shown in Supporting Information Table S7. The covalent bonds lengths in solid state are shorter compared to the bonds lengths computed in gas phase. The shortening of bonds in the experimental XRD geometry was also observed in organic compound by Katele et al[43] and was attributed to the arrangement of molecules within the crystalline solid and incorrect proton positions designation in the XRD experiment. The chloride and nitrate oxygen atoms interact with nearby ammonia groups by means of N—H···O and N—H···Cl hydrogen interactions that create hydrogen bonding chains throughout the crystal structure. The nitro and nitrito atoms groups form IHBs with nearby ammonia hydrogens in the $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ cations. The shortest Cl···H—N bonds and O···H—N bonds formed by nitrate were found to be about 2.3 and 2.1 Å in optimized **A** and **B**. The O···H—N and N···HN IHBs in $[Co(NH_3)_5NO_2]^{2+}$ and $[Co(NH_3)_5ONO]^{2+}$ averaged 2.2 Å in optimized **A** and **B** geometries. A concerted action of ionic interactions between positive charge and negative charge and the hydrogen bonds are suspected to regulate intramolecular conversion of **A** into **B** upon photoirradiation.

To probe the strength of the hydrogen bonding in the crystal structures, the AIM descriptors (Supporting Information Figure S5C,D) and the energy charge transfer E(2) were analyzed at B3LYP-D3/LanL2DZ level. Table 8 lists the AIM descriptors of the N—H···Cl, N—H···O, and

TABLE 8 Atoms in molecules (AIM) electron density (and its Laplacian in parentheses) at the bond critical point, both in atomic units, for the noncovalent interactions present in A and B unit crystals, and natural bond orbital parameters E(2) in kcal/mol (donors are chloride and nitrate and N—H acceptor) at B3LYP-D3/LanL2DZ

Properties	A	B
Cl· · ·H—N	0.01484 (0.04346))	0.01457 (0.04309)
$O(NO_3)$· · ·H—N	0.01548 (0.06758)	0.01659 (0.06331)
$O(NO_2)$· · ·H—N (IHB)	0.00830 (0.03750)	0.00554 (0.02067)
N(ONO)· · ·H—N (IHB)	—	0.01274 (0.06085)
E(2) (Cl· · ·N—H)	3.55	2.69
E(2) ((NO_3)O· · ·N—H)	2.01	1.40

N—H· · ·N bonds in **A** and **B**. The electron density at the bond critical points (bcp) between NO_3 and the N—H bond of ammonia is 0.01547 in **A** (0.01659 in **B**) while that along Cl· · ·N—H is 0.01484 in **A** (0.01457 in **B**) with the Laplacians of 0.06758 and 0.04346 (0.06331 and 0.04309 in **B**), respectively. The bcp between nitrate and its neighboring NH_3 showed greater electron density compared to that of the corresponding chloride. All computed hydrogen bond interactions listed in Table 8 had positive Laplacians, indicative of electron depletion at different bcps between the inorganic anions and cationic complexes, which may have arisen from the predominantly electrostatic interactions in the crystals. The NBO values of E(2) calculated using B3LYP-D3 with the LanL2DZ basis set represent an energetic assessment of the charge transfer from the lone pairs of chloride or those of the oxygen of nitrate anion toward the σ^* antibonding orbitals of H—N. According to the data listed in Table 8, chloride gives the larger E(2) values of ~3.55 and ~2.69 kcal/mol for **A** and **B,** respectively, compared to the corresponding values of 2.01 and 1.40 kcal/mol for the analogous charge transfer from nitrito groups. This result agrees with the NBO charge distributions listed in Supporting Information Table S8. Hence, as one can expect, crystals of **A** have higher-energy charge transfers than those of **B** owing to the strong electrostatic interactions between cationic and anionic species.

Supporting Information Figure S6 shows the high-laying occupied valence molecular orbitals and the lowest unoccupied orbitals of **A** and **B** in their singlet states. Excepting the HOMOs and LUMOs that have high metal d character, most of the higher filled molecular orbitals (eg, HOMO-1, HOMO-2, and HOMO-3) are localized on the chloride and nitro groups (Supporting Information Figure S6). $[Co(NH_3)_5NO_2](NO_3)Cl$ tetramer has a HOMO-LUMO gap of 7.47 eV which is 0.08 eV larger than that of *endo*- and *exo*-$[Co(NH_3)_5ONO](NO_3)Cl$ tetramers. The BEs per $[Co(NH_3)_5NO_2](NO_3)Cl$ molecule computed at M062X/6-31+G(d,p) in tetramer (dimer) amount to be −32.93, −32.44, and −32.33 eV for nitro, *endo*-nitrito, and *exo*-nitrito forms (−23.42, −23.67, and −22.68 eV for the dimers). The molecular orbital energy levels of these tetramers are collected in Supporting Information Table S9. The values of orbitals are close to each other but one can notice a small perturbation in molecular orbital levels created by the conversion of nitro in nitrito.

4 | CONCLUSIONS

The present work focuses on the interactions of chloride and nitrate with three linkage isomers of $[Co(NH_3)_5NO_2]^{2+}$ using ab initio (MP2 and CCSD(T)) and various DFT methods (B3LYP-D3, M062X, M06L, and wB97XD) at the 6-31+G(d,p) and LanL2DZ levels in the gas and solid phases. More precisely, various positions of attack by anions (*cis*, *trans*, and axial) and how they influence the stabilities of complexes and chemical bonding of $[Co(NH_3)_5NO_2](NO_3)Cl$ complexes were studied. Understanding the influence of inorganic anions in solid complexes of $[Co(NH_3)_5NO_2](NO_3)Cl$ linkage isomers and their chemical bonding is of great importance in the design of new photosalient materials. Our DFT and ab initio calculations led to the following conclusions:

1. Chloride interacts more strongly with $[Co(NH_3)_5NO_2]^{2+}$ linkage isomers than nitrate and prefers to be in the *trans* position.
2. $[Co(NH_3)_5NO_2](NO_3)Cl$ is thermodynamically more stable than $[Co(NH_3)_5ONO](NO_3)Cl$.
3. Interactions in $[Co(NH_3)_5NO_2](NO_3)Cl$ (**A**) and $[Co(NH_3)_5ONO](NO_3)Cl$ (**B**) crystals are essentially of ionic character and are assisted by weak hydrogen bonds.
4. Isomerization of **A** into **B** is suspected to involve an elusive intermediate chemical component.
5. Chloride and nitrate inorganic anions might act as catalysts by forming strong electrostatic bonds with the $[Co(NH_3)_5NO_2]^{2+}$ isomers, thereby lowering the energy of activation of the isomerization of the $[Co(NH_3)_5NO_2](NO_3)Cl$ complex *in vacuo* compared to the $[Co(NH_3)_5NO_2]^{2+}$ isomerization.
6. PES calculations from Frank-Condon geometry of $[Co(NH_3)_5NO_2](NO_3)Cl$ computed using CASSCF at 6-31G(d) and LanL2DZ level of theory illustrate that CI plays an important in the photoisomerization mechanism. Electronically excited S_1 of the nitro-complex relaxes through the

(S_1/S_o) CI, and undergoes a nitro-nitrito isomerization to generate the nitro and *endo*-nitrito complex. Nevertheless, *endo*-nitrito is expected to transform in *exo*-nitrito isomer on the ground state PES as both nitrito-complexes have similar relative stability.

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